

RAF-VLA: Representation Alignment with the Future for End-to-End Autonomous Driving

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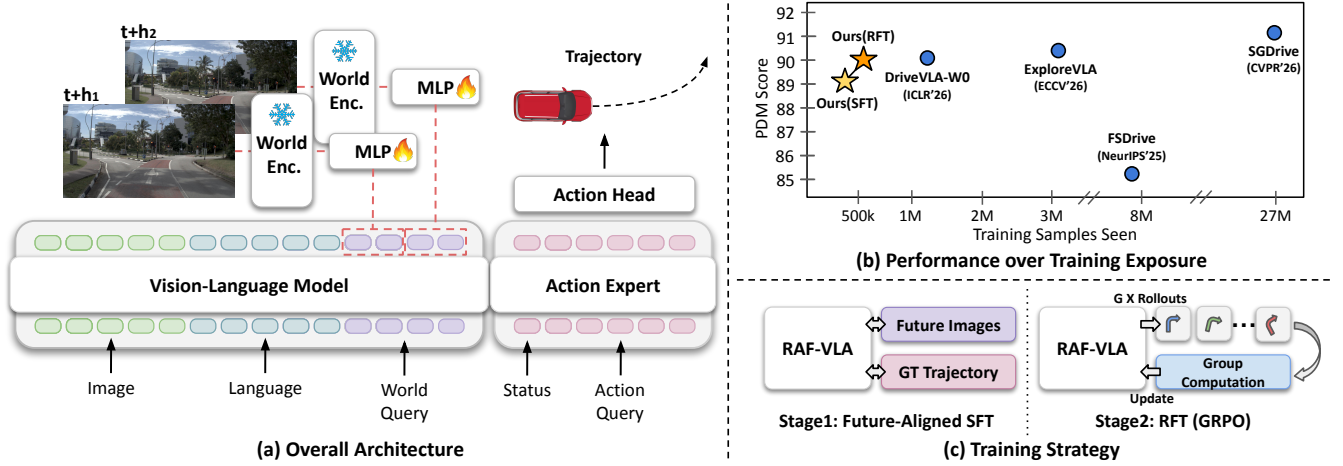


Fig. 1. Overview of RAF-VLA. (a) Overall architecture of RAF-VLA, where future-frame representations directly guide the policy’s hidden states to shape planning-relevant representations. (b) RAF-VLA achieves competitive planning performance with substantially fewer training samples seen than recent VLA planners that learn to generate future scenes. (c) Two-stage training strategy consisting of Future-Aligned SFT followed by GRPO-based RFT.

Abstract—Recent Vision-Language-Action (VLA) models for autonomous driving have incorporated world modeling by predicting future driving scenes alongside driving actions, demonstrating strong planning performance. Future driving scenes are utilized as dense supervision, encouraging the policy to learn rich internal representations useful for planning. However, these World-Modeling VLAs rely on explicit future generation to learn such representations, thereby introducing two key limitations: additional training burden and inference latency. To address these limitations, we propose RAF-VLA (Representation Alignment with the Future), a VLA-based autonomous driving framework that shapes planning-relevant internal representations through direct guidance from future-frame representations. RAF-VLA employs Future-Aligned Supervised Fine-Tuning, in which a straightforward regularization aligns the policy’s hidden states with future-frame representations obtained from a pretrained world encoder while learning driving actions. This simple alignment allows RAF-VLA to avoid the training burden and inference latency associated with future generation. Extensive experiments on the NAVSIM benchmark show that RAF-VLA achieves competitive planning performance against state-of-the-art VLA planners with substantially fewer training samples seen. Moreover, RAF-VLA incurs only 3.8% training overhead and a negligible 1 ms inference overhead.

I. INTRODUCTION

Vision-Language-Action (VLA) models have emerged as a promising approach to end-to-end autonomous driving [1] by fine-tuning Vision-Language Models (VLMs) pretrained

on Internet-scale data as driving policies [2], [3]. These models leverage VLMs’ world knowledge and reasoning capabilities to map visual observations and navigation commands to driving actions. Early VLA approaches directly predicted actions from current observations [4], [5], [6], [7], whereas subsequent studies introduced textual reasoning before action prediction to harness VLMs’ reasoning capabilities [8], [9], [10], [11], [12], [13]. However, these approaches compress continuous visual information into discrete language, which can obscure fine-grained spatial and temporal cues important for planning. Recent VLA approaches have incorporated world modeling into policy learning, predicting future driving scenes as images, depth, or occupancy alongside action prediction [14], [15], [16], [17], [18], [19], [20]. These World-Modeling VLAs utilize future driving scenes that reflect spatiotemporal structure and driving dynamics as dense supervision. Such future supervision encourages the policy to learn rich internal representations useful for planning, and World-Modeling VLAs have demonstrated strong planning performance.

However, explicit future generation in existing World-Modeling VLAs introduces two key limitations: additional training burden and inference latency. First, these methods jointly learn autoregressive generation, diffusion-based generation, or dense future-scene prediction alongside action prediction, which requires optimizing these additional generative objectives [15], [16], [17], [18], [19]. Second, predicting future scenes at inference requires additional steps, including sequential decoding or iterative denoising,

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imposing considerable latency overhead on real-time driving systems [15], [16], [18], [19]. Some methods bypass future generation during inference to avoid this overhead; however, the additional training burden remains [17].

To address these limitations, we reconsider the reliance on explicit future generation when leveraging future-scene supervision to learn planning-relevant representations. Prior studies on generative modeling show that while generative objectives can induce useful internal representations that transfer to downstream tasks [21], [22], [23], [24], [25], [26], [27], such internal representations can be shaped more directly through guidance from target representations [28], [29], [30], [31]. In driving policy learning, where the ultimate goal is planning rather than future generation, these observations motivate exploiting future-scene representations to directly shape the policy’s internal representations without a generation objective. To this end, we propose **RAF-VLA (Representation Alignment with the Future)**, a VLA-based autonomous driving framework that incorporates future-scene supervision at the representation level. RAF-VLA employs **Future-Aligned Supervised Fine-Tuning**, in which the policy learns driving actions while a straightforward regularization aligns its hidden states with future-frame representations. This simple alignment enables RAF-VLA to learn representations useful for planning without the training burden and inference latency of future generation.

We show that RAF-VLA achieves competitive planning performance against state-of-the-art VLA planners on the NAVSIM benchmark, with substantially fewer training samples seen than existing World-Modeling VLAs. Controlled comparisons and qualitative results further support the effectiveness of the proposed approach. Future-Aligned SFT incurs only a 3.8% training overhead, while latency measurements show that RAF-VLA introduces a negligible 1 ms inference overhead. The main contributions of this work are summarized as follows:

- We propose RAF-VLA, a VLA-based autonomous driving framework that leverages future-scene supervision at the representation level, enabling the driving policy to directly learn planning-relevant internal representations.
- RAF-VLA performs Future-Aligned Supervised Fine-Tuning with a straightforward regularization on action learning that aligns the policy’s hidden states with future-frame representations, thereby avoiding the training burden and inference latency of future generation.
- Experiments on the NAVSIM benchmark demonstrate competitive planning performance against state-of-the-art VLA planners, with substantially fewer training samples seen, low training overhead, and negligible inference overhead.

II. RELATED WORK

A. VLA for Autonomous Driving

Vision-Language-Action (VLA) models fine-tune pre-trained VLMs as action policies, leveraging their world knowledge and reasoning capabilities to improve generalization [3], [32], [33], [34]. This paradigm has also shown

promise in end-to-end autonomous driving, mapping visual observations and navigation commands to driving actions to address long-tail and safety-critical scenarios [2]. Early approaches focused on directly predicting driving actions from observations [4], [5], [6], [7], while subsequent studies introduced textual reasoning before action prediction to further harness VLMs’ reasoning capabilities, improving planning performance and generalization [8], [9], [10], [11], [12], [13]. Nevertheless, these approaches rely on autoregressive language-token decoding, increasing inference latency. Generating high-quality reasoning traces also incurs additional annotation costs [35]. Moreover, symbolically compressing visual information into language can obscure spatiotemporal relationships and omit fine-grained visual cues important for planning [15].

Recent studies have incorporated world modeling into VLA policy learning by predicting future driving scenes alongside actions [14], [15], [16], [17], [18], [19]. These World-Modeling VLAs utilize future scenes that reflect spatiotemporal structure and driving dynamics as dense supervision to shape useful internal representations for planning and have demonstrated strong performance [17], [18], [19]. DriveVLA-W0 [17] generates future images through autoregressive visual-token prediction or a diffusion-based generation module, while ExploreVLA [19] jointly generates future RGB and depth images through masked-token prediction. SGDrive [18] predicts future occupancy, agent states, and driving goals using task-specific prediction heads. However, exploiting future-scene supervision through future generation introduces additional training burden and inference overhead. This entails optimizing additional objectives for future generation, often also requiring dedicated training stages. Inference-time generation adds latency through autoregressive decoding or iterative denoising. Our work instead leverages future-scene supervision directly at the representation level, enabling the driving policy to learn planning-relevant internal representations without future generation.

B. Representation Learning in Generative Models

Prior studies have shown that generative objectives can induce useful internal representations for downstream tasks beyond generation [21], [22], [24], [25], [26]. More recent studies demonstrate that such representations can be shaped through direct guidance from target representations. REPA [28], [29] aligns internal diffusion features with representations from pretrained visual encoders to improve training efficiency, while iREPA [30] highlights the importance of spatial structure in target representations. ARRA [31] extends this guidance to autoregressive models. Whereas these methods pair representation guidance with generative objectives, our work uses representation guidance alone to shape planning-relevant representations for driving policies. Specifically, we align the VLA’s hidden representations with future-frame representations alongside action prediction. This leverages future-scene supervision without future generation, thereby avoiding its associated training burden and inference overhead.

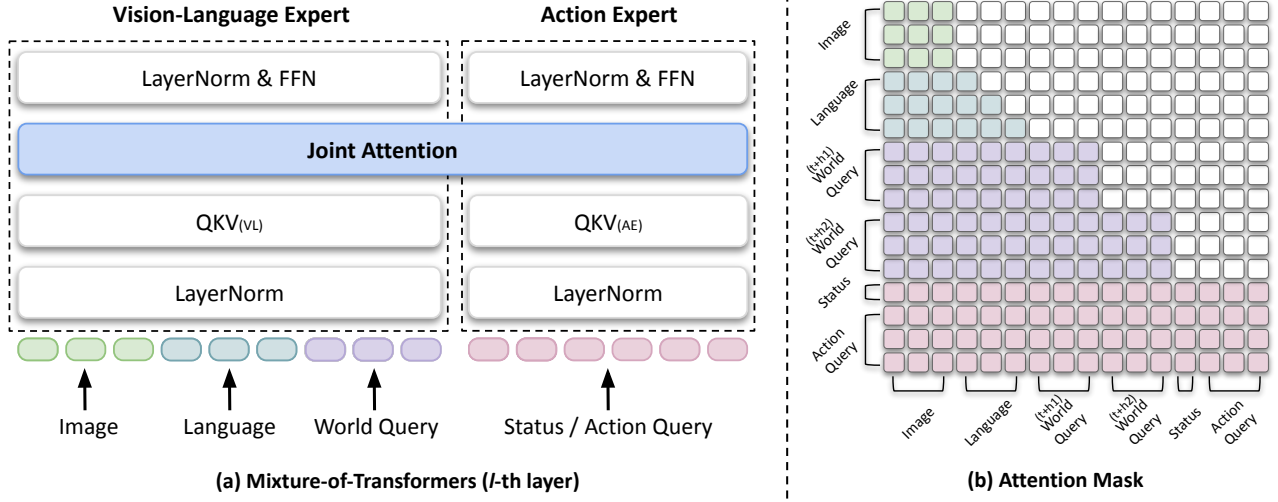


Fig. 2. Mixture-of-Transformers architecture. (a) Joint attention between the two experts. (b) Attention mask across token groups.

III. METHOD

RAF-VLA leverages future-scene supervision at the representation level to directly guide the shaping of planning-relevant representations within the driving policy. In this section, we first formulate the driving task in Sec. III-A and describe the overall architecture in Sec. III-B. We then introduce Future-Aligned Supervised Fine-Tuning (SFT) in Sec. III-C, which augments action learning with a straightforward regularization that aligns the policy’s hidden states with future-frame representations. Finally, we describe GRPO-based reinforcement fine-tuning (RFT) in Sec. III-D to further optimize the policy using planning rewards.

A. Problem Formulation

1) *Input*: At time step t , the driving policy π_θ takes a front-view image sequence, the current navigation instruction, and the driving status as input. Let $\mathbf{I}_t = \{I_{t-T}, \dots, I_t\}$ denote the front-view image sequence including the current frame I_t , where T is the observation-history length. The current navigation instruction is denoted by ℓ_t . The driving status $\mathbf{s}_t$ consists of recent ego poses and a one-hot encoded current navigation command. We denote the driving input at time step t by $q_t = (\mathbf{I}_t, \ell_t, \mathbf{s}_t)$.

2) *Output*: Given the input q_t , the driving policy π_θ predicts an ego trajectory consisting of future waypoints:

$$\hat{\tau}_t = \pi_\theta(q_t) = \{\hat{\mathbf{p}}_{t+h}\}_{h=1}^H. \quad (1)$$

Here, $\hat{\tau}_t$ denotes the predicted ego trajectory, H is the number of future steps in the planning horizon, and $h \in \{1, \dots, H\}$ is the planning-step index. Each predicted ego pose is defined as $\hat{\mathbf{p}}_{t+h} = (\hat{x}_{t+h}, \hat{y}_{t+h}, \hat{\theta}_{t+h})$, where $\hat{x}_{t+h}$ and $\hat{y}_{t+h}$ denote the predicted ego position at the h -th future step, and $\hat{\theta}_{t+h}$ denotes the corresponding heading.

B. Overall Architecture

As shown in Fig. 1(a), RAF-VLA adopts a Mixture-of-Transformers (MoT) architecture comprising a Vision-

Language Expert and a lightweight Action Expert [17], [33]. The two experts interact through layer-wise joint attention, enabling the Action Expert to leverage the Vision-Language Expert’s multimodal context for planning.

1) *Vision-Language Expert*: The Vision-Language Expert is built on a pretrained VLM to provide multimodal driving context. Its inputs, the front-view image sequence $\mathbf{I}_t$ and navigation instruction ℓ_t , are encoded by the VLM’s vision encoder and tokenizer, respectively. We append learnable world queries after the image and language tokens as representation slots for future alignment (Sec. III-C). The resulting sequence is processed to form contextualized multimodal representations.

2) *Action Expert*: The Action Expert uses a similar transformer block structure to the Vision-Language Expert, with a smaller hidden dimension. Its inputs comprise a set of learnable action queries and a status token obtained by projecting the driving status $\mathbf{s}_t$ through a multilayer perceptron (MLP). The expert leverages the Vision-Language Expert’s multimodal representations through joint attention to form action representations for trajectory prediction.

3) *Joint Attention*: As illustrated in Fig. 2(a), the two experts compute their own query, key, and value matrices at each transformer layer l . These matrices share a common attention embedding dimension and are concatenated along the token sequence dimension to form the inputs to joint attention:

$$\mathbf{Q}^l = [\mathbf{Q}_{\text{VL}}^l; \mathbf{Q}_{\text{AE}}^l], \mathbf{K}^l = [\mathbf{K}_{\text{VL}}^l; \mathbf{K}_{\text{AE}}^l], \mathbf{V}^l = [\mathbf{V}_{\text{VL}}^l; \mathbf{V}_{\text{AE}}^l]. \quad (2)$$

Here, the subscripts VL and AE denote the Vision-Language Expert and Action Expert, respectively, and the semicolon denotes concatenation. As shown in Fig. 2(b), the Action Expert applies full bidirectional attention among its status token and action queries. Attention from the Action Expert to the Vision-Language Expert is permitted, while attention in the opposite direction is masked. The resulting attention output is then split and passed back to the corresponding

expert. Following prior work [17], [36], we use a query-based action head with multiple trajectory anchors to decode the final action representations into an ego trajectory. The corresponding trajectory prediction loss is denoted by $\mathcal{L}_{\text{act}}$ [17].

C. Future-Aligned Supervised Fine-Tuning (SFT)

Future-Aligned SFT augments action learning with a straightforward regularization that aligns the policy’s hidden states with future-frame representations. This regularization directly guides the policy to learn planning-relevant representations without a separate generation objective.

1) *World Query*: World queries serve as representation slots for future-frame supervision. Let $\mathcal{H}_F \subseteq \{1, \dots, H\}$ denote the future planning steps selected for alignment. We assign K learnable world queries to each $h \in \mathcal{H}_F$. We denote their concatenated hidden states after processing by the Vision-Language Expert as $\mathbf{Z}_{t,h} \in \mathbb{R}^{Kd}$, where each hidden state is d -dimensional.

As shown in Fig. 2(b), world queries can attend to the image and language tokens. We apply blockwise causal attention to the world queries, with full bidirectional attention among the K queries assigned to each horizon, while masking attention to later-horizon queries.

2) *Representation Alignment with the Future*: As illustrated in Fig. 1(a), a frozen pretrained world encoder f extracts target representations from future frames I_{t+h} . These representations are used solely as alignment targets during training; neither the future frames nor their extracted representations are provided as inputs to the policy. A trainable MLP projector g_ϕ maps the world-query hidden states $\mathbf{Z}_{t,h}$ to the target feature space. We align the projected hidden states with the corresponding future-frame representations by minimizing their mean squared error:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{H}_F|} \sum_{h \in \mathcal{H}_F} \|f(I_{t+h}) - g_\phi(\mathbf{Z}_{t,h})\|_2^2. \quad (3)$$

3) *Training Objective*: We augment the trajectory prediction objective with the alignment regularization:

$$\mathcal{L}_{\text{SFT}} = \mathcal{L}_{\text{act}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}}, \quad (4)$$

where λ_{align} controls the regularization weight. The world encoder f remains frozen during SFT.

The alignment regularization serves as training-time guidance to shape planning-relevant representations within the policy. The projector’s outputs are used solely to compute the alignment loss. After Future-Aligned SFT, the world encoder and alignment projector are removed, while the learned world queries remain within the Vision-Language Expert. The Action Expert attends directly to their contextualized hidden states through joint attention for trajectory prediction.

D. Reinforcement Fine-Tuning (RFT)

After Future-Aligned SFT, we perform reinforcement fine-tuning (RFT) using Group Relative Policy Optimization (GRPO) [37] (Fig. 1(c)), following the standard training

strategy of recent VLA-based driving methods. GRPO improves the policy by comparing multiple candidate trajectories under the same driving context. In this stage, future-frame supervision is no longer used, and the policy is optimized with planning rewards.

Given the driving input q_t , the old policy $\pi_{\theta_{\text{old}}}$ samples a group of G candidate outputs $\mathcal{O} = \{o_1, \dots, o_G\}$, where $o_i \sim \pi_{\theta_{\text{old}}}(\cdot | q_t)$ denotes a sampled ego trajectory defined as in Eq. (1). The current policy π_θ is then optimized using the normalized group-relative advantage A_i by maximizing the following objective:

$$\mathcal{J}(\theta) = \mathbb{E}_{q_t, \mathcal{O}} \left[\frac{1}{G} \sum_{i=1}^G (\mathcal{J}_i - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})) \right], \quad (5)$$

$$\mathcal{J}_i = \min(c_i A_i, \text{clip}(c_i, 1 - \epsilon, 1 + \epsilon) A_i), \quad (6)$$

$$A_i = \frac{r_i - \text{mean}(\{r_j\}_{j=1}^G)}{\text{std}(\{r_j\}_{j=1}^G)}. \quad (7)$$

Here, $\mathcal{J}_i$ denotes the clipped surrogate objective, and $c_i = \frac{\pi_\theta(o_i | q_t)}{\pi_{\theta_{\text{old}}}(o_i | q_t)}$ is the probability ratio between the current and old policies. The group-relative advantage A_i is computed by normalizing the planning reward r_i using the mean and standard deviation of rewards within the sampled group. The hyperparameter ϵ controls clipping, while β weights the KL penalty between the current policy and the SFT reference policy π_{ref} .

The planning reward r_i is computed using the NAVSIM Predictive Driver Model Score (PDMS) [38], which evaluates trajectory quality across multiple driving criteria, including collision avoidance, drivable-area compliance, progress, time-to-collision, and comfort.

IV. EXPERIMENTS

A. Dataset and Metrics

1) *Dataset*: We conduct extensive experiments on the NAVSIM v1 and NAVSIM v2 benchmarks [38], [48] to evaluate the performance of end-to-end autonomous driving models. NAVSIM is built on OpenScene [49], which provides real-world sensor observations and annotated driving scenes. The benchmark filters out near-trivial driving cases to emphasize challenging scenarios with a higher proportion of non-trivial maneuvers. We train RAF-VLA on the *navtrain* split and evaluate on the *navtest* split.

2) *Metrics*: For NAVSIM v1, we evaluate planning performance using the Predictive Driver Model Score (PDMS):

$$\text{PDMS} = \text{NC} \times \text{DAC} \times \frac{5\text{EP} + 5\text{TTC} + 2\text{C}}{12}, \quad (8)$$

where No At-Fault Collision (NC), Drivable Area Compliance (DAC), Ego Progress (EP), Time-to-Collision (TTC), and Comfort (C) constitute the evaluation metrics.

For NAVSIM v2, we use the Extended Predictive Driver Model Score (EPDMS):

$$\begin{aligned} \text{EPDMS} = & \text{NC} \times \text{DAC} \times \text{DDC} \times \text{TLC} \\ & \times \frac{5\text{EP} + 5\text{TTC} + 2\text{LK} + 2\text{HC} + 2\text{EC}}{16}, \end{aligned} \quad (9)$$

TABLE I
COMPARISON ON NAVSIM V1 WITH CLOSED-LOOP METRICS. ABBREVIATIONS: SENSORS: 1x CAM: FRONT-VIEW CAMERA ONLY; Nx CAM: N SURROUND-VIEW CAMERAS; L: LiDAR. †: BEST-OF- N ($N = 6$) STRATEGY FOLLOWING [10], [17], [19].

Method	Venue	Sensors	NC↑	DAC↑	EP↑	TTC↑	C.↑	PDMS↑
UniAD [39]	CVPR'23	6x Cam	97.8	91.9	78.8	92.9	100.0	83.4
TransFuser [40]	TPAMI'23	3x Cam + L	97.7	92.8	79.2	92.8	100.0	84.0
LAW [41]	ICLR'25	1x Cam	96.4	95.4	81.7	88.7	99.9	84.6
Epona [42]	ICCV'25	1x Cam	97.9	95.1	80.4	93.8	99.9	86.2
Hydra-MDP [36]	arXiv'24	3x Cam + L	98.3	96.0	78.7	94.6	100.0	86.5
DiffusionDrive [43]	CVPR'25	3x Cam + L	98.2	96.2	82.2	94.7	100.0	88.1
WoTE [44]	ICCV'25	3x Cam + L	98.5	96.8	81.9	94.9	99.9	88.3
DriveLaW [45]	CVPR'26	1x Cam	99.0	97.1	81.3	96.7	100.0	89.1
<i>Textual-Reasoning VLA</i>								
AutoVLA [10]	NeurIPS'25	3x Cam	98.4	95.6	81.9	98.0	99.9	89.1
ReCogDrive [11]	ICLR'26	3x Cam	98.2	97.8	83.5	95.2	99.8	89.6
AdaThinkDrive [9]	ICRA'26	1x Cam	98.4	97.8	84.4	95.2	100.0	90.3
AutoVLA† [10]	NeurIPS'25	3x Cam	99.1	97.1	87.6	97.1	100.0	92.1
AdaThinkDrive† [9]	ICRA'26	1x Cam	99.1	98.8	87.9	97.2	100.0	93.0
<i>World-Modeling VLA</i>								
DrivingGPT [14]	ICCV'25	1x Cam	98.9	90.7	79.7	94.9	95.6	82.4
FSDrive [15]	NeurIPS'25	3x Cam	98.2	93.8	80.1	93.3	99.9	85.1
PWM [16]	NeurIPS'25	1x Cam	98.6	95.9	81.8	95.4	100.0	88.1
DriveVLA-W0 [17]	ICLR'26	1x Cam	98.7	99.1	83.3	95.3	99.3	90.2
ExploreVLA [19]	ECCV'26	1x Cam	98.8	98.4	83.5	96.5	99.9	90.4
SGDrive [18]	CVPR'26	1x Cam	98.6	97.8	85.8	96.2	100.0	91.1
DriveVLA-W0† [17]	ICLR'26	1x Cam	99.3	97.4	88.3	97.0	99.9	93.0
ExploreVLA† [19]	ECCV'26	1x Cam	99.4	98.9	88.3	98.3	99.7	93.7
RAF-VLA (Ours)	–	1x Cam	98.0	98.5	85.2	93.9	99.7	90.0
RAF-VLA† (Ours)	–	1x Cam	99.0	99.6	89.8	97.3	100.0	94.2

TABLE II
NAVSIM V2 RESULTS WITH EXTENDED CLOSED-LOOP METRICS USING FUTURE-ALIGNED SFT.

Method	NC↑	DAC↑	DDC↑	TLC↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑
TransFuser [40]	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7
VADv2 [46]	97.3	91.7	98.2	99.9	77.6	92.7	66.0	100.0	97.4	76.6
Hydra-MDP [36]	97.5	96.3	98.3	99.9	80.1	93.0	65.5	100.0	97.4	79.8
Hydra-MDP++ [47]	97.2	97.5	99.4	99.6	83.1	96.5	94.4	98.2	70.9	81.4
ReCogDrive [11]	98.3	95.2	99.5	99.8	87.1	97.5	96.6	98.3	86.5	83.6
DiffusionDrive [43]	98.2	95.9	99.4	99.8	87.5	97.3	96.8	98.3	87.7	84.5
DriveVLA-W0 [17]	98.5	99.1	98.0	99.7	86.4	98.1	93.2	97.9	58.9	86.1
SGDrive [18]	98.6	94.3	99.5	99.9	86.0	97.9	96.1	98.3	85.9	86.2
RAF-VLA (Ours)	97.5	98.2	99.4	99.8	86.9	97.0	92.7	98.0	65.7	86.2

which additionally includes Driving Direction Compliance (DDC), Traffic Light Compliance (TLC), Lane Keeping (LK), History Comfort (HC), and Extended Comfort (EC). Higher values indicate better performance for all metrics.

B. Implementation Details

We use Qwen2.5-VL-7B-Instruct as the Vision-Language Expert [50], a lightweight transformer with an embedding dimension of 1024 as the Action Expert, and a pretrained encoder from the Cosmos world model [51] to extract target representations from future frames. The alignment projector is a two-layer MLP with a GELU activation [52]. For Future-

Aligned SFT, we train for 4 epochs with AdamW, a global batch size of 144, and a learning rate of 1×10^{-5} with cosine decay, while freezing the VLM vision encoder and world encoder. Unless otherwise specified, alignment uses the 1 s and 4 s horizons with $K = 4$ world queries per horizon and $\lambda_{\text{align}} = 1$. For RFT, we perform GRPO for 1.5 epochs with a global batch size of 288, a learning rate of 2×10^{-6} , group size $G = 8$, and KL coefficient $\beta = 0.04$; the world encoder and alignment projector are removed, while the VLM vision encoder, world queries, and action head remain frozen. Both stages are trained on $4 \times \text{H200}$ GPUs, taking approximately 11 hours for SFT and 7 hours for RFT.



Fig. 3. Qualitative comparison of trajectory predictions on the NAVSIM *navtest* split. From left to right: front-camera input, vanilla SFT without alignment, Future-Aligned SFT, and Future-Aligned SFT followed by GRPO-based RFT (black: GT, red/orange/green: prediction from each model).

TABLE III

ABLATION OF FUTURE-REPRESENTATION ALIGNMENT UNDER SFT AND RFT. BOLD VALUES INDICATE THE BEST RESULT FOR EACH METRIC.

ID	Alignment	SFT	RFT	NC $\uparrow$	DAC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	C. $\uparrow$	PDMS $\uparrow$
1		✓		97.86	98.58	81.01	93.03	99.52	87.92
2		✓	✓	97.42	98.39	84.31	92.38	99.51	88.85
3	✓	✓		98.04	98.71	82.99	93.60	99.79	89.13
4	✓	✓	✓	97.98	98.46	85.21	93.86	99.73	90.03

TABLE IV

INFERENCE LATENCY WITH AND WITHOUT ALIGNMENT. INFERENCE TIME DENOTES THE AVERAGE TIME REQUIRED TO PREDICT 4-SECOND TRAJECTORIES ON THE NAVSIM *v1 navtest* SPLIT.

Alignment	Latency (s) $\downarrow$	Δ
×	0.152	–
✓	0.153	+0.001

C. Main Results

1) *Performance Comparison*: Table I compares RAF-VLA with state-of-the-art (SOTA) methods on NAVSIM *v1*. Using only a single front-view camera, RAF-VLA achieves a PDMS of 90.0, demonstrating competitive planning performance against SOTA VLA-based planners. As shown in Fig. 1(b), this performance is achieved with substantially fewer training samples seen than existing World-Modeling VLAs. The total training samples seen for each method are estimated from its reported configurations, official implementations, and official dataset splits. With the best-of- N strategy, the PDMS further increases to 94.2, achieving the highest overall performance, together with the best DAC and EP scores of 99.6 and 89.8, respectively. We further evaluate the Future-Aligned SFT model on NAVSIM *v2* using the

extended metric. As shown in Table II, it achieves an EPDMS of 86.2, matching the best reported SFT result.

2) *Effectiveness of Future-Aligned SFT*: Table III evaluates the effect of the regularization for representation alignment in Future-Aligned SFT, both before and after reinforcement fine-tuning. For the vanilla SFT baseline, we remove the entire future-alignment mechanism, including the associated world queries. Future-Aligned SFT improves the PDMS from 87.92 to 89.13 (+1.21) over SFT without alignment. This improvement incurs only a 3.8% increase in training time, from 10.56 h (42.2 H200 GPU-hours) for vanilla SFT to 10.96 h (43.8 H200 GPU-hours) for Future-Aligned SFT, including online extraction of future representations. Importantly, this gain is retained after reinforcement fine-tuning. The model initialized from Future-Aligned SFT achieves a PDMS of 90.03, compared with 88.85 for the model initialized from SFT without alignment (+1.18). These results show that the benefit introduced during Future-Aligned SFT persists through subsequent RFT.

Fig. 3 provides a qualitative comparison of trajectory predictions from SFT without alignment, Future-Aligned SFT, and Future-Aligned SFT followed by RFT. SFT without alignment produces trajectories that poorly follow the lane consistent with the navigation command and deviate from the ground-truth trajectory. They also fail to make sufficient progress toward the intended direction. Future-Aligned SFT corrects these failures, yielding trajectories that better follow the intended maneuver, while subsequent RFT further refines the predicted paths. The quantitative and qualitative results consistently support the effectiveness of Future-Aligned SFT.

3) *Inference Overhead*: As shown in Table IV, Future-Aligned SFT adds only 0.001 s (1 ms) of latency for predicting a 4-second trajectory, resulting in negligible inference overhead. Inference latency is measured on a single NVIDIA

TABLE V
ABLATION OF FUTURE-ALIGNMENT HORIZONS ON NAVSIM v1.

ID	1s	2s	3s	4s	NC↑	DAC↑	EP↑	TTC↑	C.↑	PDMS↑
1					97.86	98.58	81.01	93.03	99.52	87.92
2	✓				98.40	98.50	82.33	93.91	99.81	88.93
3				✓	98.30	98.67	82.39	93.78	99.76	88.97
4	✓	✓			97.80	98.48	82.32	93.47	99.53	88.60
5	✓			✓	98.04	98.71	82.99	93.60	99.79	89.13
6	✓	✓	✓	✓	97.88	98.53	82.52	93.13	99.79	88.62

TABLE VI
ABLATION OF THE QUERY COUNT K ON NAVSIM v1.

ID	K	NC↑	DAC↑	EP↑	TTC↑	C.↑	PDMS↑
1	2	98.14	98.36	82.94	93.21	99.78	88.76
2	4	98.04	98.71	82.99	93.60	99.79	89.13
3	8	97.94	98.63	82.30	93.25	99.76	88.64
4	16	97.97	98.63	82.49	93.03	99.79	88.65

H200 GPU with a batch size of 1. This is because Future-Aligned SFT directly shapes policy representations through alignment during training, leaving only a few learned world queries to be processed during the prefill stage at inference. No additional scene prediction or generation is required.

D. Ablation Studies

Unless otherwise specified, all ablation studies follow the implementation details in Sec. IV-B and are conducted under the SFT setting.

1) *Effect of Future Alignment Horizons*: Table V compares different future horizons used for representation alignment. Using either the 1s or 4s future representation improves the planning performance over the vanilla SFT baseline, with the 4s horizon corresponding to the end of the planning horizon. Among the evaluated configurations, jointly using the 1s and 4s horizons achieves the highest PDMS of 89.13, while adding intermediate horizons does not provide further improvement. We therefore use the 1s and 4s horizons as the default configuration.

2) *Effect of the Number of World Queries*: Table VI ablates the number of world queries assigned to each future horizon. Among the evaluated settings, $K = 4$ achieves the highest PDMS of 89.13, while increasing the number of world queries beyond four does not provide further improvement. We therefore use four world queries per horizon as the default configuration.

V. CONCLUSIONS

In this work, we introduced RAF-VLA, a VLA-based autonomous driving framework that leverages future-scene supervision at the representation level without future generation. RAF-VLA performs Future-Aligned Supervised Fine-Tuning, which aligns the policy’s hidden states with future-frame representations, thereby providing direct guidance for shaping planning-relevant internal representations during action learning. Experiments on NAVSIM v1 and v2 demonstrate competitive planning performance with substantially

fewer training samples seen, low training overhead, and negligible inference overhead. Controlled comparisons and qualitative results provide further evidence for the effectiveness of our proposed approach. Future work will explore alternative target encoders and projection designs. We also plan to extend future representation alignment to multi-view inputs and richer scene modalities.

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